Symbiosis Institute of Technology, Pune

Question Bank

Name of the Department: Artificial Intelligence & Machine Learning

Name of the Course: Natural Language Processing (TEE7101) **Course Code:** 0701260505

Class: B.Tech **Credit:** 03 **Batch:** 2024-2028 **Semester-** V

Name of the Faculty: Dr. Sucheta and Prof. Mayur

Unit 1: Introduction to Language & Preprocessing (CO1)

1. Apply tokenization, stemming, and lemmatization on a given text and compare the outputs.
2. Analyze the impact of different preprocessing techniques on text classification performance.
3. Given a sentence, perform POS tagging and interpret the grammatical structure.
4. Apply Named Entity Recognition on a paragraph and categorize entities.
5. Compare normalization techniques and justify their use in different NLP tasks.
6. Analyze challenges faced in preprocessing multilingual text data.
7. Evaluate the importance of preprocessing in improving NLP model accuracy.
8. Given raw text data, design a preprocessing pipeline and justify each step.

Unit 2: Feature Extraction & Language Models (CO2)

1. Apply Bag of Words and TF-IDF on a dataset and compare the feature vectors.
2. Analyze the differences between CountVectorizer and TF-IDF with examples.
3. Given a sentence corpus, compute unigram and bigram probabilities.
4. Apply Naïve Bayes for text classification and interpret the results.
5. Analyze the limitations of N-gram models in language modeling.
6. Compare generative and probabilistic models for NLP tasks.
7. Evaluate smoothing techniques in handling unseen words.
8. Given a dataset, select an appropriate feature extraction method and justify your choice.

Unit 3: Word Embeddings & Neural Networks (CO3)

1. Apply Word2Vec or GloVe embeddings on a dataset and interpret word similarities.

2. Analyze the difference between one-hot encoding and word embeddings.
3. Given a set of words, visualize embeddings and interpret semantic relationships.
4. Apply neural network concepts to classify text data.
5. Analyze how context is captured in word embeddings.
6. Compare Word2Vec and GloVe models with suitable examples.
7. Evaluate the effectiveness of embeddings over traditional feature extraction methods.
8. Given a problem, choose an embedding technique and justify your decision.

Unit 4: Recurrent Neural Networks (CO4)

1. Apply RNN for sequence prediction and interpret the output.
2. Analyze the vanishing gradient problem in RNNs.
3. Compare LSTM and GRU architectures with respect to performance and complexity.
4. Given a sequence dataset, design an RNN-based model.
5. Analyze the role of memory cells in LSTM.
6. Evaluate the effectiveness of RNNs over traditional language models.
7. Given a problem, select between RNN, LSTM, and GRU with justification.
8. Analyze limitations of RNNs in long sequence modeling.

Unit 5: Machine Translation & Advanced NLP (CO5)

1. Apply Seq2Seq model for a simple translation task and interpret results.
2. Analyze the working of encoder-decoder architecture in machine translation.
3. Compare traditional machine translation techniques with neural approaches.
4. Evaluate the performance of transformer-based models over RNNs.
5. Given a translation problem, design an appropriate NLP solution.
6. Analyze challenges in multilingual machine translation.
7. Evaluate recent advancements in NLP models for language tasks.
8. Given a real-world scenario, justify the choice of NLP model and approach.